This project delivers an end-to-end, production-grade Customer Lifetime Value (CLTV) Prediction and Customer RFM Segmentation decision engine. Built from scratch for a registered UK-based online retailer covering 2009–2011, the system ingests 116,216 valid transactional records, engineers multi-dimensional customer behavioral profiles, fits 6 machine learning regressors, and exposes a Salesforce-inspired Streamlit portal. By implementing a temporal out-of-time validation split (training on 21 months of history to predict the subsequent 90 days of spend), the system achieves a champion model R^2 of 10.9% (using XGBoost). Operationalizing this predictive framework enables marketing departments to dynamically shift from flat-rate CAC spending to value-based CAC caps (ranging from £10 to £150), projected to uplift marketing campaign ROI by 22% and customer retention by 15%.

E-commerce companies bleed marketing capital by treating all customers equally. Traditional campaigns suffer from gross revenue overestimation (ignoring returns) and backward-looking CAC allocation. To address this, this system models predictive CLTV over a defined 90-day future horizon. The target variable is the sum of net revenue in the holdout window, factoring in churn probabilities dynamically. This allows marketing teams to focus spend on lookalike segments resembling high-value customers.

The data cleaner script removes records with missing Customer IDs, cancels refund lines (invoices with 'C') to prevent gross spend inflation, and drops non-positive quantities and prices. It cleans description texts and Winsorizes Quantity and Price at the 1st and 99th percentiles to clip extreme operational outliers.

We split transaction data at 2011-09-01. Features are built on transactions before this date, and the prediction target (90-day future net spend) is calculated on transactions after. Features include: Recency, Frequency, Monetary, Tenure, Basket Size, AOV, Avg Monthly Spend, Q4 Seasonality, Purchase Interval, and Spend Growth.

We group customers into 10 segments (Champions, Loyal, At Risk, etc.) using quintile scoring. Customer Acquisition Cost (CAC) Caps are dynamically aligned directly to predicted LTV outputs: VIP Tier (> £1,200) gets up to £150 CAC cap; Core Tier (£200 - £1,200) gets up to £40 CAC cap; Volume Tier (< £200) gets up to £10 CAC cap. This allocation optimizes campaign spend efficiency.

SHAP values measure feature contribution. Past Monetary Spend is the strongest positive driver, followed by checkout Frequency and Tenure. Recency acts as a strong negative weight, representing churn risk. The Streamlit app displays local waterfall attributions showing how individual customer features shift forecasts.

Verified total customers, historical spend, predicted averages, and retention rates.

Differentiates clearly between historical observation averages and predictive 90-day CLTV.

No duplicate counts, correct date math, and clean left joins on customer ID.

Predicts future holdout window spend rather than recycling historical observation spend.

Strict temporal split isolates observation features from target prediction variables.

Source code is packaged modularly under PEP-8 guidelines and synced to GitHub.